

Teaching Statement

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Teaching Philosophy

My teaching philosophy is grounded in three principles: **conceptual clarity**, **hands-on practice**, and **student-centered mentoring**. Computer science education should help students understand the fundamental ideas behind computing while also giving them the ability to implement, test, debug, and communicate solutions to real problems. I view teaching not only as content delivery, but also as a process of building confidence, discipline, curiosity, and independent problem-solving ability.

Across more than seven years of university-level teaching experience, I have taught students with diverse preparation levels in Computer Science and Information and Communication Technology programs. This experience has shaped my belief that effective teaching requires a balance between theory and practice. Students learn best when abstract concepts are connected to code, systems, projects, case studies, and real-world applications.

Courses Taught and Teaching Breadth

My teaching experience covers foundational, systems, security, and AI-related areas of computer science. I have taught or supported undergraduate courses including:

- Structured Programming Language
- Object-Oriented Programming Language
- Data Structures
- Algorithms
- Operating Systems
- Information Security
- Artificial Intelligence
- Machine Learning
- Deep Learning

This teaching breadth allows me to connect lower-level programming and systems concepts with advanced AI and security topics. For example, in programming and data structures courses, I emphasize algorithmic thinking, debugging, modular design, and code readability. In operating systems and information security, I connect theory with practical examples involving processes, memory, access control, vulnerabilities, and secure design. In AI, machine learning, and deep learning, I focus on both mathematical foundations and implementation skills, including model evaluation, data quality, overfitting, reproducibility, and ethical use of AI.

Instructional Approach

My classroom approach combines structured explanation with active learning. I usually begin by motivating a topic through a practical problem, then introduce the core concept, work through examples, and finally guide students toward independent implementation or analysis. I encourage students to ask questions, compare alternative solutions, and explain their reasoning in their own words.

For programming and algorithmic courses, I use step-by-step coding demonstrations, small exercises, debugging sessions, and project-based assignments. For AI and machine learning courses, I emphasize experimental thinking: students should know how to prepare data, choose baselines, train models, evaluate results, interpret errors, and report findings responsibly. For security-related courses, I emphasize responsible computing, threat awareness, secure software practices, and the importance of protecting users, data, and systems.

I also believe assessment should measure both understanding and application. My preferred assessment structure includes quizzes or short conceptual checks, programming assignments, laboratory tasks, project milestones, presentations, and final reports. This approach helps students develop not only technical competence but also communication and professional skills.

Mentoring and Student Supervision

Student mentoring has been one of the most important parts of my academic career. I have supervised **56 undergraduate projects** and **15 thesis students** across Port City International University and Comilla University. These projects covered software development, database systems, web and mobile applications, data

mining, machine learning, health informatics, security, and management information systems.

My supervision style is structured and supportive. I help students refine a feasible project idea, identify the problem statement, review related work, design the system or methodology, implement the solution, evaluate results, and prepare written documentation. I encourage students to treat projects as opportunities to learn research habits: reading, planning, experimentation, version control, presentation, and reflection.

I have also supported student development through academic advising and programming activities. As coach of the CoU “Unpredictable 3207” ACM ICPC team, I helped students develop problem-solving confidence and competitive programming discipline. These experiences strengthened my commitment to mentoring students both inside and outside the classroom.

Inclusive and Student-Centered Teaching

I believe that a strong classroom should be rigorous, respectful, and inclusive. Students come with different levels of confidence, programming exposure, mathematical preparation, and language background. My role is to maintain high academic standards while creating a learning environment where students feel encouraged to participate and improve.

To support diverse learners, I use multiple forms of explanation: diagrams, code examples, pseudocode, real-world analogies, and stepwise problem decomposition. I also encourage collaborative learning through discussion, peer feedback, and project teamwork, while making expectations and grading criteria clear. When students struggle, I try to identify whether the challenge is conceptual, technical, or confidence-related, and then guide them with targeted feedback.

Teaching Development and Professional Growth

I have regularly participated in workshops, seminars, and professional development activities related to outcome-based education, curriculum design, research methodology, scholarly writing, machine learning, artificial intelligence, cybersecurity, information security, data mining, natural language processing, software engineering, student mentoring, undergraduate research supervision, and research proposal writing. These activities have helped me improve my teaching practice and remain current with developments in computer science education and research.

My ongoing Ph.D. training has further strengthened my teaching. Working on trustworthy AI, quantum machine learning, medical image analysis, and secure/adversarial machine learning allows me to bring current research examples into the classroom. I aim to help students see how foundational courses such as programming, algorithms, operating systems, and security connect to modern AI systems, biomedical computing, and emerging technologies.

Future Teaching Goals

My future teaching goals are to continue developing as an educator who can teach both core computer science courses and advanced AI/security courses. I am prepared to teach programming, object-oriented design, data structures, algorithms, operating systems, information security, artificial intelligence, machine learning, deep learning, secure machine learning, data mining, software engineering, and biomedical AI-related topics.

I also aim to develop project-based learning opportunities where students can work on real datasets, reproducible experiments, secure software systems, and AI applications with social and biomedical relevance. My long-term goal is to help students become technically strong, ethically aware, and capable of contributing to research, industry, and society.

Summary

Teaching is central to my academic identity. My experience as a former Assistant Professor, Lecturer, Senior Lecturer, project supervisor, thesis mentor, and Ph.D. researcher has prepared me to teach a wide range of computer science courses and guide students in both foundational and research-oriented learning. I am committed to clear instruction, rigorous practice, inclusive mentoring, and continuous improvement as a teacher.